

There are two n -element arrays of integers, A and B . Permute them into some A' and B' such that the relation $A'[i] + B'[i] \geq k$ holds for all i where $0 \leq i < n$.

There will be q queries consisting of A , B , and k . For each query, return YES if some permutation A' , B' satisfying the relation exists. Otherwise, return NO.

Example

$A = [0, 1]$

$B = [0, 2]$

$k = 1$

A valid A' , B' is $A' = [1, 0]$ and $B' = [0, 2]$: $1 + 0 \geq 1$ and $0 + 2 \geq 1$. Return YES.

Function Description

Complete the *twoArrays* function in the editor below. It should return a string, either YES or NO.

twoArrays has the following parameter(s):

- *int k*: an integer
- *int A[n]*: an array of integers
- *int B[n]*: an array of integers

Returns

- *string*: either YES or NO

Input Format

The first line contains an integer q , the number of queries.

The next q sets of 3 lines are as follows:

- The first line contains two space-separated integers n and k , the size of both arrays A and B , and the relation variable.
- The second line contains n space-separated integers $A[i]$.
- The third line contains n space-separated integers $B[i]$.

Constraints

- $1 \leq q \leq 10$
- $1 \leq n \leq 1000$
- $1 \leq k \leq 10^9$
- $0 \leq A[i], B[i] \leq 10^9$

Sample Input

STDIN	Function
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2	$q = 2$
3 10	$A[]$ and $B[]$ size $n = 3$, $k = 10$
2 1 3	$A = [2, 1, 3]$
7 8 9	$B = [7, 8, 9]$
4 5	$A[]$ and $B[]$ size $n = 4$, $k = 5$
1 2 2 1	$A = [1, 2, 2, 1]$
3 3 3 4	$B = [3, 3, 3, 4]$

Sample Output

YES

NO

Explanation

There are two queries:

1. Permute these into $A' = [1, 2, 3]$ and $B' = [9, 8, 7]$ so that the following statements are true:

- $A[0] + B[1] = 1 + 9 = 10 \geq k$
- $A[1] + B[1] = 2 + 8 = 10 \geq k$
- $A[2] + B[2] = 3 + 7 = 10 \geq k$

2. $A = [1, 2, 2, 1]$, $B = [3, 3, 3, 4]$, and $k = 5$. To permute A and B into a valid A' and B' , there must be at least three numbers in A that are greater than 1.